Layer–Time Geometry

Layer–Time Geometry

How Language Models Process Information

Agus Sudjianto
H2O.ai

Aijun Zhang
Wells Fargo

An undergraduate-friendly companion to
Layer–Time Geometry of Transformer Computation

Preface

Who this book is for

This book is for undergraduate data science students who want to understand *what happens inside a transformer* when it processes text. You know what a neural network is. You have used PyTorch or TensorFlow. You may have fine-tuned a language model. But you have probably never asked: what does the internal state of the model *look like*, mathematically, as it moves from layer to layer?

That is the question this book answers.

What you need

- **Linear algebra**: matrix multiplication, eigenvalues, SVD. If you can multiply two matrices and know what an eigenvector is, you are ready.
- **Basic statistics**: mean, variance, PCA, ANOVA (at the "I've seen it in a stats class" level).
- **Python/PyTorch**: enough to load a model and run a forward pass.
- **Calculus**: partial derivatives and gradients. Nothing beyond what you see in machine learning courses.

You do *not* need differential geometry, measure theory, geometric algebra, or operator theory. The research monograph that this book is based on uses all of those. This book translates the key ideas into language you already speak.

How this book works

Every chapter follows the same pattern:

1. **The big question**: what are we trying to understand?
2. **Analogy**: a concrete, everyday comparison.

3. **The idea**: the core concept, in plain language.
4. **The math you need**: only what is necessary, explained step by step.
5. **Python code**: how to compute it yourself.
6. **What we found**: empirical results from real models.
7. **So what?**: why this matters for your work.

Colored boxes throughout the text serve specific purposes:

- **Why Should I Care?** — motivates the topic.
- **Key Idea** — the central takeaway.
- **Analogy** — an intuitive comparison.
- **Try It Yourself** — hands-on exercises.
- **Common Pitfall** — mistakes to avoid.
- **What the Data Shows** — empirical findings.
- **Under the Hood** — optional deeper math (skip on first reading).

Relationship to the research monograph

This book is a companion to *Layer–Time Geometry of Transformer Computation: Kernels, Operators, and Dependency in Language Models* (Sudjianto and Zhang, 2026), which develops the full mathematical framework with proofs, theorems, and formal definitions. When you are ready for the complete treatment, that is where to go. This book gives you the intuition to read it.

The `ltg` library

Every chapter in this book includes runnable code using the `ltg` (Layer-Time Geometry) Python library. This is a student-friendly API that wraps the research-grade `layer_time` package. Install it with pip:

Listing 1: Install the layer-time package

```
pip install layer-time
```

This gives you access to two APIs:

- `import ltg` — the student-friendly API used in this book
- `from layer_time import LayerTimeAnalyzer` — the full research API

The API is designed around three verbs:

Listing 2: The entire ltg workflow in four lines

```python
import ltg

model = ltg.load_model("Qwen/Qwen2.5-7B")      # load once
result = ltg.analyse("What is 2 + 3?", model)  # analyse any prompt
result.summary()                                 # see what happened
result.plot_all(prefix="my_analysis")            # generate all plots
```

Every `AnalysisResult` object carries the full geometric decomposition (whitened states, curvature map, polar decomposition, dependency profile) and provides one-line plotting methods. You can compare prompts, run control experiments, and diagnose failures—all without touching the underlying linear algebra.

Companion Jupyter notebooks

Every chapter has a companion Jupyter notebook in the `tutorials/` directory. These notebooks contain all the code from the chapter—ready to run, with visualisations and exercises. Install the tutorial dependencies with:

Listing 3: Install tutorial extras

```
pip install layer-time[tutorials]
```

Notebook	Chapter
`ch1_opening_the_black_box.ipynb`	Ch. 1: Opening the Black Box
`ch2_whitening.ipynb`	Ch. 2: Standardizing the Representation
`ch3_kernels.ipynb`	Ch. 3: Measuring Similarity: Kernels
`ch4_polar_decomposition.ipynb`	Ch. 4: How Information Flows
`ch5_curvature.ipynb`	Ch. 5: Where the Action Is: Curvature
`ch6_experiments.ipynb`	Ch. 6: Testing What Matters: DOE
`ch7_dependency.ipynb`	Ch. 7: What Each Layer Contributes
`ch8_reasoning_memory_control.ipynb`	Ch. 8: Reasoning, Memory, and Control
`ch9_diagnosing_failures.ipynb`	Ch. 9: Practical Applications

Each notebook is self-contained: load the model once at the top, then work through the cells sequentially. GPU access (CUDA) is recommended but not strictly required—CPU will work for short prompts, just more slowly.

— *The Authors*

Contents

Preface **i**

I What Is Going On Inside? **1**

1 Opening the Black Box **3**
1.1 The question 3
1.2 Hidden states: the model's internal workspace 4
1.3 The four big ideas 5
1.4 What you will be able to do 6
1.5 Hands-on: your first analysis in 60 seconds 6

2 Standardizing the Internal Representation **9**
2.1 The problem with raw hidden states 9
2.2 The solution: PCA-based whitening 10
2.3 Why this matters: metric consistency 10
2.4 Python implementation 11
2.5 Using the ltg library 12
2.6 What we skip on first reading 13

3 Measuring Similarity: Kernels on the Grid **15**
3.1 What is a kernel? 15
3.2 The layer kernel: which layers are similar? 16
3.3 The temporal kernel: which tokens are similar? 16
3.4 What the kernels reveal: three computational phases 16
3.5 Hands-on: visualizing the three phases 17

4 How Information Flows: Operators and Polar Decomposition **19**

4.1 The layer transition operator 19
4.2 Polar decomposition: rotation + stretch 20
4.3 What rotation and stretching mean 21
4.4 Measuring the stretch: eigenvalues of P 21
4.5 Hands-on: exploring the polar decomposition with ltg 22

5 Where the Action Is: Curvature 25
5.1 The idea of curvature on a grid 25
5.2 Computing curvature on the layer–time grid 26
5.3 What curvature tells us 27
5.4 Hands-on: curvature maps with ltg 27

II What Do the Experiments Say? 29

6 Testing What Matters: Experimental Design 31
6.1 The experimental setup 31
6.2 Three models 32
6.3 The results: five key findings 32
6.4 Hands-on: your own mini-DOE 34

7 What Each Layer Contributes: Dependency 37
7.1 The idea 37
7.2 Computing dependency 38
7.3 Dependency profiles 38
7.4 Hands-on: dependency profiles with ltg 39

III The Big Picture 41

8 Reasoning, Memory, and Control 43
8.1 Defining reasoning geometrically 43
8.2 Defining memory 44
8.3 Defining control 44
8.4 Hands-on: running the control experiment yourself 45
8.5 Hands-on: measuring functional memory 47

9 Practical Applications: Diagnosing Model Failures 49
9.1 Application 1: Detecting when a model ignores context 49

9.2 Application 2: Detecting hallucination types 50
9.3 Application 3: Steering model behavior 52

10 What We Learned and What's Next **55**
10.1 The validated findings 55
10.2 What this book did not cover 56
10.3 Open problems for you to work on 56
10.4 The one thing to remember 57

Glossary **60**

Further Reading **61**

Part I

What Is Going On Inside?

CHAPTER 1

Opening the Black Box

Companion notebook: `tutorials/ch1_opening_the_black_box.ipynb` — run the code from this chapter interactively.

Why Should I Care?

You use language models every day—for code completion, summarisation, question answering. But when a model hallucinates, how do you diagnose the failure? When a model ignores your instructions, where in the computation did things go wrong? To answer these questions, you need to see inside.

1.1 The question

A transformer language model takes a sequence of tokens and processes them through a stack of layers. At each layer, every token has a **hidden state**—a high-dimensional vector that encodes everything the model "knows" about that token at that point in the computation.

The central question of this book is:

What is the geometric structure of these hidden states,
and what does it tell us about the model's computation?

1.2 Hidden states: the model's internal workspace

Analogy

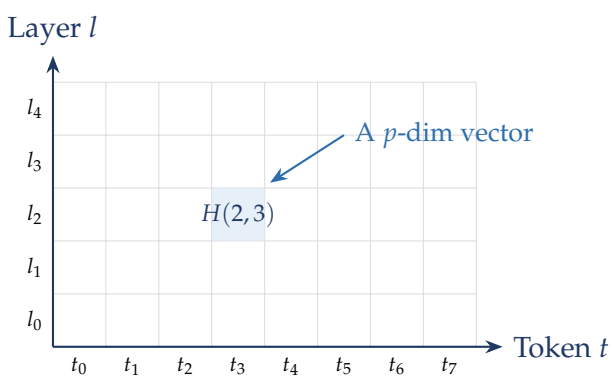
Think of a transformer like a factory assembly line. Raw materials (tokens) enter at one end. At each station (layer), workers (attention heads and feed-forward networks) modify the product. The hidden state at each station is like a detailed spec sheet describing the product's current state—its shape, color, material, intended use. By the time the product reaches the end of the line, the spec sheet contains everything needed to produce the final output.

This book is about reading those spec sheets and understanding what the workers are doing at each station.

Concretely, if a model has L layers and processes T tokens, then at layer l and token position t, the model holds a vector:

$$H(l, t) \in \mathbb{R}^p$$

where p is the model's hidden dimension (e.g., $p = 4096$ for a 7B parameter model). The collection of all these vectors forms a two-dimensional grid:



We call this the **layer–time grid**. "Layer" is the vertical axis (depth in the network). "Time" is the horizontal axis (position in the sequence). Every cell holds a p-dimensional vector.

Key Idea

The hidden states of a transformer form a **field** on a two-dimensional grid. The geometry of this field—how vectors change across layers and across tokens—reveals the model's internal computation.

1.3 The four big ideas

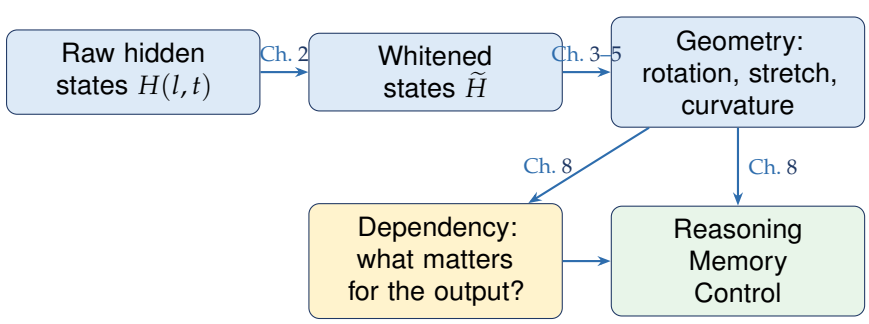
This book develops four main ideas, each building on the last:

1. **Standardize the vectors** (Chapter 2). Raw hidden states have correlated, differently-scaled dimensions. We fix this with PCA-based whitening, like standardizing variables before running a regression.

2. **Measure how vectors change** (Chapters 3–5). We decompose the layer-to-layer transformation into a rotation (direction change) and a stretch (magnitude change), then measure curvature—how much the grid's geometry twists.

3. **Run controlled experiments** (Chapter 6). We use factorial experimental design (DOE) to test which input properties actually affect the geometry: task type? reasoning difficulty? prompt length?

4. **Connect geometry to reasoning** (Chapter 8). We add a new observable—*dependency*—that measures how much each layer's hidden state matters for the model's final output. This lets us define reasoning, memory, and control in geometric terms.



1.4 What you will be able to do

By the end of this book, you will be able to:

- Extract and whitened hidden states from any Hugging Face transformer.
- Compute the polar decomposition of layer transitions and interpret what rotation and stretching mean.
- Measure curvature and identify where in the network non-trivial computation happens.
- Design and analyze factorial experiments on geometric quantities.
- Compute dependency profiles that show which layers matter most for the model's output.
- Diagnose specific failure modes (hallucination, instruction-ignoring, reasoning errors) using geometric signatures.

1.5 Hands-on: your first analysis in 60 seconds

Listing 1.1: Complete first analysis using the ltg library

```python
import ltg

# Step 1: Load the model (takes ~30 seconds, only do this once)
model = ltg.load_model("Qwen/Qwen2.5-7B")

# Step 2: Analyse a prompt (takes ~10 seconds)
result = ltg.analyse("What is the square root of 144?", model=model)

# Step 3: See the summary
result.summary()
# Output:
#   Tokens: 9  |  Layers: 28  |  Whitened dim: 256
#   Mean curvature: 0.0847
#   Peak layer: 26 (of 27)
#   Final-3-layer share: 42.3%
#   ...

# Step 4: Generate all plots
result.plot_all(prefix="ch1_first")
# Creates: ch1_first_curvature.png, ch1_first_kernel.png,
#          ch1_first_polar.png, ch1_first_dependency.png
```

Try It Yourself

Run the code above. Then try two more prompts:

- A factual question: `"Who wrote Romeo and Juliet?"`
- A reasoning problem: `"If all cats are animals and Whiskers is a cat, what is Whiskers?"`

Compare the three summaries. Which has the highest curvature? Which has the lowest dependency entropy? Do the differences surprise you?

CHAPTER 2

Standardizing the Internal Representation

Companion notebook: `tutorials/ch2_whitening.ipynb` — run the code from this chapter interactively.

Why Should I Care?

If you compare hidden states at different layers without standardizing them first, you are comparing apples and oranges. Whitening is to hidden-state analysis what z-scoring is to statistical analysis: it removes scale differences and correlations so that the geometry you measure is real, not an artefact of the coordinate system.

2.1 The problem with raw hidden states

A transformer's hidden state vector has p dimensions (typically 2048–8192). These dimensions are *not* independent:

- Some dimensions carry much more variance than others.
- Many dimensions are correlated.
- The scale changes from layer to layer.

If you directly compute distances between hidden states at different layers, the distance is dominated by whichever dimensions happen to have the largest variance—

not necessarily the most informative ones.

Analogy

Imagine measuring the similarity between cities using two features: population (in millions) and average temperature (in degrees). Population ranges from 0.01 to 30; temperature ranges from -20 to 45. Without standardisation, Euclidean distance is almost entirely determined by population, and temperature barely matters. The same problem occurs in hidden states, but with thousands of dimensions.

2.2 The solution: PCA-based whitening

The fix is a standard statistical tool: **whitening via PCA**.

Step 1: Center. Subtract the mean hidden state $\bar{H}$ computed across all layers and tokens for a given input.

Step 2: Compute the covariance. The $p \times p$ covariance matrix Σ captures how dimensions co-vary.

Step 3: PCA projection. Take the top k eigenvectors $V_k = [v_1, \ldots, v_k]$ of Σ, with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_k > 0$.

Step 4: Whiten. Define the whitening matrix $W = V_k \operatorname{diag}(\lambda_1^{-1/2}, \ldots, \lambda_k^{-1/2})$ and compute:

$$\widetilde{H}(l,t) = \big(H(l,t) - \bar{H}\big)\, W \quad \in \mathbb{R}^k$$

The result: $\widetilde{H}$ has uncorrelated dimensions with unit variance. We use $k = 256$ throughout—this retains the meaningful variation while making computation tractable.

Key Idea

After whitening, Euclidean distance in the k-dimensional space equals Mahalanobis distance in the original p-dimensional space. This means the geometry we measure respects the model's internal metric, not an arbitrary coordinate choice.

2.3 Why this matters: metric consistency

The core theoretical result (proven formally in the research monograph) is:

Metric Consistency Theorem (informal). After whitening, the covariance of the hidden states is the identity matrix. All subsequent geometric operations—decompositions, distances, curvatures—are invariant to the original coordinate system of the model.

In plain language: whitening lets us compare geometry *across layers, across prompts, and across models* in a principled way. Without it, we would be measuring artefacts.

2.4 Python implementation

Listing 2.1: Whitening hidden states from a Hugging Face model

```python
import torch
from transformers import AutoModelForCausalLM, AutoTokenizer

def extract_and_whiten(model, tokenizer, text, k=256):
    """Extract whitened hidden states from a transformer."""
    inputs = tokenizer(text, return_tensors="pt").to(model.device)

    with torch.no_grad():
        out = model(**inputs, output_hidden_states=True)

    # Stack all layers: shape (L+1, T, p)
    H = torch.stack(out.hidden_states)     # includes layer 0
    H = H[1:]                               # skip embedding layer
    L, T, p = H.shape

    # Step 1: Centre
    H_flat = H.reshape(-1, p)               # (L*T, p)
    H_mean = H_flat.mean(dim=0)
    H_centred = H_flat - H_mean

    # Step 2-3: PCA via SVD
    U, S, Vt = torch.linalg.svd(H_centred, full_matrices=False)
    V_k = Vt[:k].T                          # (p, k)
    eigenvalues = (S[:k] ** 2) / (L * T - 1)

    # Step 4: Whiten
    W = V_k * (1.0 / eigenvalues.sqrt())    # (p, k)
    H_whitened = H_centred @ W              # (L*T, k)
    H_whitened = H_whitened.reshape(L, T, k)

    return H_whitened  # shape: (L, T, k)
```

2.5 Using the ltg library

The `ltg.analyse()` function performs whitening automatically. But you can inspect the whitened states directly:

Listing 2.2: Inspecting whitened hidden states with ltg

```python
import ltg
import numpy as np

model = ltg.load_model("Qwen/Qwen2.5-7B")
result = ltg.analyse("The capital of France is", model=model)

# Access the whitened hidden states
H_w = result.H_whitened  # shape: (L, T, k)
print(f"Shape: {H_w.shape}")
# e.g., Shape: (28, 7, 256)

# Verify whitening: covariance should be close to identity
H_flat = H_w.reshape(-1, H_w.shape[2])  # (L*T, k)
cov = np.cov(H_flat.T)                   # (k, k)

# Check: diagonal should be ~1, off-diagonal should be ~0
print(f"Mean diagonal:     {np.diag(cov).mean():.4f}")   # ~1.0
print(f"Mean off-diagonal: {np.abs(cov - np.diag(np.diag(cov))).mean
    ():.6f}")  # ~0.0
print(f"Max off-diagonal:  {np.abs(cov - np.diag(np.diag(cov))).max()
    :.4f}")
```

Try It Yourself

Exercise 2.1: Run the code above and verify the covariance is approximately I_k.

Exercise 2.2 (Whitening dimension): Try $k = 32, 64, 128, 256$. For each, compute the curvature profile and check whether the shape changes. At what k does the profile stabilise?

```python
# Hint: use the core library directly for custom k
import layer_time_geometry as core

for k in [32, 64, 128, 256]:
    H_raw = core.extract_hidden_states(model.hf_model, model.
        tokenizer,
                                       "The capital of France is
                                           ", "cuda")
```

```python
    H_np = H_raw[1:].cpu().numpy()
    metric = core.estimate_metric(H_np, n_components=k)
    H_w = core.whiten(H_np, metric)
    Omega = core.curvature(H_w)
    print(f"k={k:3d}: mean_curv={Omega.mean():.4f}, "
          f"peak_layer={Omega.mean(axis=1).argmax()}")
```

2.6 What we skip on first reading

The research monograph proves several additional properties: that whitening is the unique linear transformation achieving metric consistency, that the effective rank of the covariance determines the intrinsic dimensionality, and that the choice of k is stable for $k \geq 128$.

Common Pitfall

Do not skip layer 0. Layer 0 is the raw token embedding before any transformer processing. Including it in the whitening distorts the geometry because it has a fundamentally different distribution. Always start from layer 1. The `ltg` library handles this automatically.

CHAPTER 3

Measuring Similarity: Kernels on the Grid

Companion notebook: `tutorials/ch3_kernels.ipynb` — run the code from this chapter interactively.

Why Should I Care?

Kernels answer the most basic question about the hidden-state grid: which layers are similar to each other? Which token positions are similar? The answers reveal the model's computational phases—setup, processing, and decision-making.

3.1 What is a kernel?

A kernel is just a similarity function. Given two objects a and b, a kernel $K(a, b)$ returns a number indicating how similar they are. You already know several kernels:

- **Dot product**: $K(a, b) = a^\top b$ — the simplest kernel.
- **Cosine similarity**: $K(a, b) = \frac{a^\top b}{\|a\|\|b\|}$ — normalized dot product.
- **RBF kernel**: $K(a, b) = \exp(-\|a - b\|^2/2\sigma^2)$ — similarity decays with distance.

We define two kernels on the layer–time grid.

3.2 The layer kernel: which layers are similar?

For each layer l, collect all T whitened token vectors into a matrix $\widetilde{H}^{(l)} \in \mathbb{R}^{T \times k}$. The **layer kernel** measures how similar two layers are:

$$K_{\text{layer}}(l, l') = \frac{1}{T} \operatorname{tr}\left(\widetilde{H}^{(l)\top} \widetilde{H}^{(l')}\right)$$

This is just the average dot product between corresponding token vectors at two layers.

Analogy

Think of each layer as a photograph of the assembly line taken from above. The layer kernel measures how much two photographs look alike. Early layers look similar to each other (setup phase), late layers look similar to each other (decision phase), and there is often a visible transition in the middle (processing phase).

3.3 The temporal kernel: which tokens are similar?

Similarly, for each token t, collect its representation across all layers: $\widetilde{H}_t \in \mathbb{R}^{L \times k}$. The **temporal kernel** measures:

$$K_{\text{time}}(t, t') = \frac{1}{L} \operatorname{tr}\left(\widetilde{H}_t^\top \widetilde{H}_{t'}\right)$$

3.4 What the kernels reveal: three computational phases

What the Data Shows

When we compute the layer kernel for typical prompts across multiple models (Qwen2.5-7B, DeepSeek-R1-7B, Qwen2.5-32B), we consistently find **three phases**:

1. **Early layers** ($l \lesssim L/4$): High mutual similarity. The model is building up basic representations.
2. **Middle layers** ($L/4 \lesssim l \lesssim 3L/4$): Rapid change. This is where the "work" happens—attention heads route information, representations diverge.
3. **Late layers** ($l \gtrsim 3L/4$): Convergence toward a prediction-ready state. Curvature peaks here.

Listing 3.1: Computing the layer kernel matrix

```
def layer_kernel(H_whitened):
```

```
    """Compute the L x L layer kernel matrix.

    Args:
        H_whitened: tensor of shape (L, T, k)
    Returns:
        K: tensor of shape (L, L)
    """
    L, T, k = H_whitened.shape
    # Reshape to (L, T*k) and compute normalised Gram matrix
    H_flat = H_whitened.reshape(L, T * k)
    K = H_flat @ H_flat.T / T
    return K
```

3.5 Hands-on: visualizing the three phases

Listing 3.2: Visualising the layer kernel to see computational phases

```
import ltg

model = ltg.load_model("Qwen/Qwen2.5-7B")
result = ltg.analyse("Explain why the sky is blue", model=model)

# The layer kernel is already computed
result.plot_layer_kernel(save_path="ch3_kernel.png")

# You will see a block-diagonal structure:
#   - Top-left block (early layers): high similarity
#   - Middle: transition zone
#   - Bottom-right block (late layers): high similarity
# The transition boundary marks where the model shifts
# from "understanding" to "deciding".
```

Try It Yourself

Exercise 3.1: Compare layer kernels for a short prompt (5 tokens) and a long prompt (50+ tokens). Does the three-phase structure hold for both?

Exercise 3.2 (Phase boundary detection): The phase boundary is where consecutive layer similarity drops. Write code to find it:

```
K = result.layer_kernel_matrix
# Consecutive similarity: K[l, l+1] for each l
consec = [K[l, l+1] for l in range(K.shape[0] - 1)]
import numpy as np
```

```
transition = np.argmin(consec)
print(f"Phase transition at layer {transition}")
```

Where does the transition occur relative to the total depth?

CHAPTER 4

How Information Flows: Operators and Polar Decomposition

Companion notebook: `tutorials/ch4_polar_decomposition.ipynb` — run the code from this chapter interactively.

Why Should I Care?

This is the chapter where the framework becomes powerful. We decompose *how* hidden states change from one layer to the next into two fundamental operations: rotation and stretching. This decomposition is the key to understanding what each layer actually does.

4.1 The layer transition operator

Between layer l and layer $l+1$, the hidden states change. We can approximate this change with a linear operator $T^{(l)}$ that best maps $\widetilde{H}^{(l)}$ to $\widetilde{H}^{(l+1)}$:

$$\widetilde{H}^{(l+1)} \approx \widetilde{H}^{(l)}\, T^{(l)}$$

We find $T^{(l)}$ by ordinary least squares—the same linear regression you learned in your first statistics course, just applied to matrices:

$$T^{(l)} = \left(\widetilde{H}^{(l)\top}\widetilde{H}^{(l)}\right)^{-1}\widetilde{H}^{(l)\top}\widetilde{H}^{(l+1)}$$

Analogy

If the hidden states at layer l are the "input data" and the hidden states at layer $l+1$ are the "output data", then $T^{(l)}$ is the best linear predictor. It tells us: what linear transformation best explains how the representation changed?

4.2 Polar decomposition: rotation + stretch

Every square matrix can be decomposed as:

$$T^{(l)} = U^{(l)}\,P^{(l)}$$

where:

- $U^{(l)}$ is an **orthogonal matrix** (rotation/reflection): it changes directions without changing magnitudes.
- $P^{(l)}$ is a **symmetric positive-definite matrix** (stretching): it scales different directions by different amounts.

This is called the **polar decomposition**, and it is analogous to writing a complex number as $z = e^{i\theta} \cdot r$ (angle + magnitude).

Key Idea

Every layer transition in a transformer does two things:

1. **Rotate** the representation (change which directions are active), captured by $U^{(l)}$.
2. **Stretch** the representation (amplify some directions, compress others), captured by $P^{(l)}$.

These are the two fundamental primitives of transformer computation.

4.3 What rotation and stretching mean

Component	What it does	Computational meaning
$U^{(l)}$ (rotation)	Rotates the coordinate axes; preserves all distances	Attention mixing: redistributes information across directions without losing or amplifying anything
$P^{(l)}$ (stretch)	Stretches some directions, compresses others	Feature selection: amplifies important directions, suppresses irrelevant ones

4.4 Measuring the stretch: eigenvalues of P

The eigenvalues $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k$ of $P^{(l)}$ tell us how much each principal direction is stretched:

- $\sigma_i > 1$: direction i is amplified (the model is paying more attention to this direction).
- $\sigma_i < 1$: direction i is compressed (the model is discarding information along this direction).
- $\sigma_i = 1$: direction i is unchanged.

The **condition number** $\kappa = \sigma_1/\sigma_k$ measures how selective the layer is. A high condition number means the layer strongly amplifies some directions while crushing others—it is making a sharp "decision".

The **effective rank** $\text{erank} = \exp(H(\hat{\sigma}))$, where H is the entropy of the normalized eigenvalue distribution, measures how many directions are actively used. Low effective rank means the layer concentrates its computation in a few directions.

Listing 4.1: Polar decomposition of layer transitions

```python
def polar_decompose_layers(H_whitened):
    """Compute polar decomposition T = U @ P for each layer.

    Args:
        H_whitened: (L, T, k) whitened hidden states
```

```
    Returns:
        Us: list of L-1 orthogonal matrices (k, k)
        Ps: list of L-1 symmetric positive-definite matrices (k, k)
    """
    L, T, k = H_whitened.shape
    Us, Ps = [], []

    for l in range(L - 1):
        H_l = H_whitened[l]        # (T, k)
        H_next = H_whitened[l+1]   # (T, k)

        # Least-squares transition operator
        T_l = torch.linalg.lstsq(H_l, H_next).solution  # (k, k)

        # Polar decomposition
        U, S, Vt = torch.linalg.svd(T_l)
        P = Vt.T @ torch.diag(S) @ Vt   # symmetric positive-definite
        U_orth = U @ Vt                   # orthogonal
        Us.append(U_orth)
        Ps.append(P)

    return Us, Ps
```

4.5 Hands-on: exploring the polar decomposition with ltg

The `ltg` library computes the polar decomposition automatically. You can visualize it and inspect individual layers:

Listing 4.2: Polar decomposition analysis with ltg

```
import ltg
import numpy as np

model = ltg.load_model("Qwen/Qwen2.5-7B")
result = ltg.analyse("If x = 3 and y = 2*x + 1, what is y?", model=
    model)

# Plot condition number and effective rank across layers
result.plot_polar(save_path="ch4_polar.png")

# Inspect a specific layer's stretch
layer = 25  # a late layer
P = result.polar_P[layer]             # (k, k) symmetric matrix
sigmas = np.linalg.svd(P, compute_uv=False)  # eigenvalues

```

```python
print(f"Layer {layer} stretch spectrum:")
print(f"  Top 5 eigenvalues:    {sigmas[:5]}")
print(f"  Bottom 5 eigenvalues: {sigmas[-5:]}")
print(f"  Condition number:     {sigmas[0]/sigmas[-1]:.1f}")
print(f"  Effective rank:       {result.effective_ranks[layer]:.1f}")
# High condition number = the layer is being very selective
# Low effective rank = computation concentrates in few directions
```

Try It Yourself

Exercise 4.1 (Rotation vs. stretch magnitude): For each layer, compute:

- $\|U^{(l)} - I\|_F$ (how much rotation deviates from identity)
- $\|P^{(l)} - I\|_F$ (how much stretching deviates from identity)

```python
for l in range(result.n_layers - 1):
    U_dev = np.linalg.norm(result.polar_U[l] - np.eye(result.k),
        'fro')
    P_dev = np.linalg.norm(result.polar_P[l] - np.eye(result.k),
        'fro')
    print(f"Layer {l:2d}: rotation={U_dev:.3f}, stretch={P_dev:.3
        f}")
```

Which layers have more rotation? Which have more stretching? Is there a pattern?

Exercise 4.2 (Compare tasks): Analyze an arithmetic prompt and a factual retrieval prompt. Compare their condition number profiles. Does the model make sharper "decisions" for one task type?

CHAPTER 5

Where the Action Is: Curvature

Companion notebook: `tutorials/ch5_curvature.ipynb` — run the code from this chapter interactively.

Why Should I Care?

Curvature tells you *where* in the layer–time grid the interesting computation happens. Zero curvature means the model could be replaced by a simple linear transformation at that point. High curvature means something genuinely non-trivial is occurring. This is the single most useful diagnostic in the framework.

5.1 The idea of curvature on a grid

In everyday life, curvature measures how much a surface bends. A flat table has zero curvature. A basketball has constant positive curvature. A saddle has negative curvature.

On the layer–time grid, curvature measures something analogous: how much the geometry "twists" when you go around a small loop.

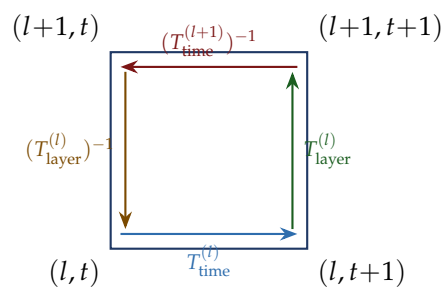
Analogy

Imagine walking around a square block in a city. You make four right turns and end up facing the same direction—zero curvature. Now imagine walking the same square

on the surface of a sphere (like walking a small square near the North Pole). After four right turns, you *do not* face the same direction—the surface's curvature rotated you. The same idea applies to hidden states: go right one token, go up one layer, go left one token, go down one layer. If you end up with a different transformation than the identity, there is curvature.

5.2 Computing curvature on the layer–time grid

Consider a small square ("plaquette") at position (l, t) on the grid:



Going around this loop gives a combined transformation:

$$\Omega(l, t) = T^{(l)}_{\text{time}} \cdot T^{(l)}_{\text{layer}} \cdot \left(T^{(l+1)}_{\text{time}}\right)^{-1} \cdot \left(T^{(l)}_{\text{layer}}\right)^{-1}$$

If the grid were "flat" (i.e., moving right then up gives the same result as moving up then right), then $\Omega(l, t) = I$. The deviation from identity *is* the curvature:

$$\text{curvature at } (l, t) = \|\Omega(l, t) - I\|_F$$

where $\|\cdot\|_F$ is the Frobenius norm (think: sum of squared entries, then square root).

Key Idea

Curvature measures **non-commutativity**: does it matter whether you go "right then up" or "up then right" on the grid? If yes, the computation at that point is genuinely nonlinear and non-trivial.

5.3 What curvature tells us

What the Data Shows

Across all models tested (three architectures, two scales), curvature consistently shows the same pattern:

- **Early layers**: Low curvature. The model is doing simple, nearly-linear processing.
- **Middle layers**: Moderate curvature. Information routing and mixing.
- **Final 3–5 layers**: Curvature spikes sharply. This is where the model commits to its prediction.

This is the most robust finding in the entire framework—it replicates across models, prompts, and tasks.

Common Pitfall

Curvature does not equal reasoning complexity. Early in this research, we conjectured that higher curvature meant harder reasoning. Controlled experiments across three models showed this is **false**: reasoning depth (number of logical steps) has no significant effect on curvature. Curvature measures *non-separable interaction* between layers and tokens—not how "hard" the model is thinking.

5.4 Hands-on: curvature maps with ltg

Listing 5.1: Curvature analysis with ltg

```python
import ltg

model = ltg.load_model("Qwen/Qwen2.5-7B")

# Analyse two prompts of different "difficulty"
r_easy = ltg.analyse("What is 2 + 2?", model=model)
r_hard = ltg.analyse(
    "If f(x) = 3x^2 - 2x + 1, what is f'(x) evaluated at x = 4?",
    model=model
)

# Plot curvature maps for both
r_easy.plot_curvature(save_path="ch5_curv_easy.png")
r_hard.plot_curvature(save_path="ch5_curv_hard.png")

# Compare them side by side
ltg.compare([r_easy, r_hard], save_path="ch5_compare.png")
```

```python
# Print summaries
print("Easy prompt:")
print(f"  Peak curvature layer: {r_easy.curvature_by_layer.argmax()}")
print(f"  Final-3-layer share:  "
      f"{r_easy.curvature_by_layer[-3:].sum()/r_easy.curvature_by_layer.sum():.1%}")
print("Hard prompt:")
print(f"  Peak curvature layer: {r_hard.curvature_by_layer.argmax()}")
print(f"  Final-3-layer share:  "
      f"{r_hard.curvature_by_layer[-3:].sum()/r_hard.curvature_by_layer.sum():.1%}")
# Key finding: the peak location is similar! Curvature does NOT
# increase with problem difficulty.
```

Try It Yourself

Exercise 5.1 (Testing the negative result): Analyze five prompts with increasing reasoning depth (1-step, 2-step, ..., 5-step). Plot their mean curvature. Does it increase? (Spoiler: it should not.)

Exercise 5.2 (Curvature across models): If you have access to a second model (e.g., `Qwen/Qwen2.5-1.5B`), compare curvature profiles on the same prompt. Does the three-phase structure persist at smaller scale?

Part II

What Do the Experiments Say?

CHAPTER 6

Testing What Matters: Experimental Design

Companion notebook: `tutorials/ch6_experiments.ipynb` — run the code from this chapter interactively.

Why Should I Care?

Anyone can compute a metric. The hard part is knowing whether the metric actually tells you something about the input. Experimental design (DOE) is the tool that separates real signal from noise. This chapter uses the same factorial designs you learn in statistics courses—applied to transformer geometry.

6.1 The experimental setup

We designed 320 prompts varying five factors:

Factor	Levels
Cognitive task	Arithmetic, Deductive Reasoning, Causal Reasoning, Linguistic Analysis, Factual Retrieval, Definition
Reasoning depth	Low (1-step), Medium (2-3 steps), High (4+ steps)
Prompt length	Short (<50 tokens), Medium, Long (>150 tokens)
Knowledge domain	Science, History, Everyday, Abstract
Robustness	Original, Paraphrased

For each prompt, we extracted whitened hidden states and computed ∼20 geometric metrics (curvature, condition number, effective rank, dependency measures, etc.). Then we ran ANOVA: which factors significantly affect which metrics?

6.2 Three models

We ran the experiment on three models to check replicability:

Model	Parameters	Layers
Qwen2.5-7B	7 billion	28
DeepSeek-R1-Distill-Qwen-7B	7 billion	28
Qwen2.5-32B	32 billion	64

6.3 The results: five key findings

What the Data Shows

R1: Task type matters, but modestly.

Cognitive task type (arithmetic vs. reasoning vs. retrieval, etc.) is *detectable* in the geometry—certain dependency metrics differ significantly across task types. But the effect sizes are small ($\eta^2 \approx 0.02$–0.05). Task type is a real signal, not the dominant driver.

What the Data Shows

R2: Surface features are not negligible.

We expected that "shallow" features like prompt length and formatting would have no effect on geometry. **Wrong.** Prompt length affects dependency horizons significantly, especially in DeepSeek-R1. Robustness condition drives the largest single effect in Qwen-7B ($\eta^2 = 0.098$ for the global curvature ratio). The geometry responds to both computational content *and* input form.

What the Data Shows

R3: Reasoning depth has no effect on geometry.

This is the cleanest and most important negative result. Across all three models, the factor "reasoning depth" (1-step vs. multi-step problems) *fails to reach significance* on any geometric metric. The one minor exception: condition number in Qwen-7B ($\eta^2 = 0.032$, $p = 0.010$).

What this means: you cannot tell how many reasoning steps a problem requires by looking at the geometry of the hidden states. The number of logical steps does not map to any measurable internal geometric quantity.

What the Data Shows

R4: Curvature concentrates in the final layers.

This finding replicates unanimously across all three models. In Qwen2.5-32B (64 layers), peak curvature is entirely in the final 3 layers. This is the most robust qualitative finding in the study.

What the Data Shows

R5: Dependency tracks task type more closely than curvature.

Dependency-based metrics (how much each layer's state matters for the output) are more sensitive to task type than curvature-based metrics. They are also more stable under paraphrase.

Finding	Qwen-7B	DS-R1	Qwen-32B
Task type detectable	yes	yes	yes
Surface features weak	**no**	**no**	**no**
Curvature = reasoning depth	**no**	**no**	**no**
Curvature in final layers	yes	yes	yes
Dependency stable	yes	yes	yes

6.4 Hands-on: your own mini-DOE

You can replicate a small version of our experiment with `ltg`:

Listing 6.1: Mini factorial experiment with ltg

```
import ltg
import numpy as np
from scipy import stats

model = ltg.load_model("Qwen/Qwen2.5-7B")

# Define prompts by task type
prompts = {
    "arithmetic": [
        "What is 17 * 23?",
        "Calculate 156 / 12",
        "What is 2^10?",
    ],
    "reasoning": [
        "If all dogs are pets and Rex is a dog, is Rex a pet?",
        "If it rains then the ground is wet. The ground is wet. Did it rain?",
        "Alice is taller than Bob. Bob is taller than Carol. Who is shortest?",
    ],
    "retrieval": [
        "What is the capital of Japan?",
        "Who painted the Mona Lisa?",
        "What year did World War II end?",
    ],
}

# Collect metrics by task type
```

```python
results_by_task = {}
for task, texts in prompts.items():
    metrics = []
    for text in texts:
        r = ltg.analyse(text, model=model, compute_dependency=True)
        metrics.append({
            'mean_curv': r.curvature_by_layer.mean(),
            'peak_layer': r.curvature_by_layer.argmax(),
            'dep_entropy': r.dep_entropy,
            'dep_h90': r.dep_horizon_90,
        })
    results_by_task[task] = metrics

# Simple one-way ANOVA on dependency entropy
groups = [
    [m['dep_entropy'] for m in results_by_task[t] if m['dep_entropy']]
    for t in prompts.keys()
]
F, p = stats.f_oneway(*groups)
print(f"ANOVA on dep_entropy: F={F:.3f}, p={p:.3f}")
print("Task type significant?" , "Yes" if p < 0.05 else "No")
```

Try It Yourself

Exercise 6.1: Expand the mini-DOE to 5 prompts per task type and add a fourth task (e.g., "definition"). Run the ANOVA. Which metric shows the strongest task-type effect?

Exercise 6.2 (Paraphrase stability): Take one prompt and create three paraphrases. Analyze all four versions. How similar are their curvature profiles? Compute the pairwise correlation.

CHAPTER 7

What Each Layer Contributes: Dependency

Companion notebook: `tutorials/ch7_dependency.ipynb` — run the code from this chapter interactively.

Why Should I Care?

Geometry tells you *what shape* the hidden states have. Dependency tells you *which parts matter*. A layer might have complex geometry but contribute nothing to the final output. Dependency distinguishes "busy work" from "real work."

7.1 The idea

After a model processes a prompt, it produces a score (log-probability) for the next token. Some layers contributed heavily to this score; others barely mattered. **Dependency** measures this contribution using gradients.

Analogy

In a relay race, some runners gain time and others lose time. The final clock time depends on all runners, but not equally. Dependency is like looking at the split times to figure out who actually made the difference.

7.2 Computing dependency

The dependency of layer l is:

$$D_l = \left\| \frac{\partial s}{\partial \widetilde{H}^{(l)}} \right\|_F$$

where s is the model's score for the next token. This is just the gradient of the output with respect to the hidden state at layer l. Layers where the gradient is large "mattered more" for the output.

Listing 7.1: Computing dependency profiles

```python
def dependency_profile(model, tokenizer, text, k=256):
    """Compute dependency D_l for each layer."""
    inputs = tokenizer(text, return_tensors="pt").to(model.device)

    # Forward pass with gradient tracking
    out = model(**inputs, output_hidden_states=True)
    hidden_states = out.hidden_states[1:]  # skip layer 0

    # Score: log-prob of the predicted next token
    logits = out.logits[0, -1, :]          # last position
    score = logits.max()                    # score of top token

    # Backward pass
    score.backward()

    # Collect gradient norms per layer
    D = []
    for hs in hidden_states:
        if hs.grad is not None:
            D.append(hs.grad.norm().item())
        else:
            D.append(0.0)

    return D  # list of length L
```

Note: the code above is simplified. In practice, you need to enable gradient computation for intermediate hidden states, which requires hooks. See the research monograph's implementation appendix for the full version.

7.3 Dependency profiles

The dependency profile $D_0, D_1, \ldots, D_{L-1}$ reveals which layers drive the output.

Useful summary statistics:

- $D_{\text{total}} = \sum_l D_l$ — total sensitivity to all layers.
- **Dependency horizon** H_α — the layer at which α% of the total dependency is reached. A low H_{90} means the model makes up its mind early.
- **Dependency entropy** $S = -\sum_l \hat{D}_l \log \hat{D}_l$ (where $\hat{D}_l = D_l / D_{\text{total}}$) — how spread out the dependency is. High entropy = many layers contribute; low entropy = a few layers dominate.

Common Pitfall

Length confound. Total dependency D_{total} is strongly correlated with sequence length ($r \approx 0.96$). Longer sequences have more hidden states for the gradient to flow through, mechanically increasing D_{total}. Always use **normalized** metrics (entropy, horizons, concentration ratios) when comparing prompts of different lengths.

7.4 Hands-on: dependency profiles with ltg

Listing 7.2: Computing and visualising dependency profiles

```python
import ltg
import numpy as np

model = ltg.load_model("Qwen/Qwen2.5-7B")

# Analyse with dependency (default: compute_dependency=True)
result = ltg.analyse(
    "The Pythagorean theorem states that a^2 + b^2 = c^2",
    model=model
)

# Plot the dependency profile
result.plot_dependency(save_path="ch7_dependency.png")

# Access the raw profile
D = result.dependency_profile
D_norm = D / D.sum()  # normalise

print(f"Total dependency:          {result.dep_total:.4f}")
print(f"Dependency entropy:        {result.dep_entropy:.3f}")
print(f"Horizon-90 (layer):        {result.dep_horizon_90}")
print(f"Final-3-layer concentration: {result.dep_concentration_final3
    :.1%}")

```

```
# Find the top-3 most important layers
top3 = np.argsort(D_norm)[-3:][::-1]
for l in top3:
    print(f"  Layer {l}: {D_norm[l]:.1%} of total dependency")
```

Try It Yourself

Exercise 7.1 (Length confound): Analyze prompts of lengths 10, 30, and 100 tokens. Plot D_{total} vs. token count. Verify the correlation. Then plot dependency *entropy* vs. token count—is the confound still present?

Exercise 7.2 (Comparing dependency and curvature): For a single prompt, overlay the normalized curvature profile and the normalized dependency profile on the same plot:

```
import matplotlib.pyplot as plt
curv_norm = result.curvature_by_layer / result.curvature_by_layer
    .sum()
dep_norm = result.dependency_profile / result.dependency_profile.
    sum()
# Curvature has L-1 entries, dependency has L entries
plt.plot(curv_norm, label='Curvature (normalised)')
plt.plot(dep_norm, label='Dependency (normalised)')
plt.xlabel('Layer'); plt.legend()
plt.savefig('ch7_curv_vs_dep.png')
```

Do they peak at the same layers? What does it mean when curvature is high but dependency is low?

Part III

The Big Picture

CHAPTER 8

Reasoning, Memory, and Control

Companion notebook: `tutorials/ch8_reasoning_memory_control.ipynb` — run the code from this chapter interactively.

Why Should I Care?

This chapter connects everything. The geometry and dependency tools from earlier chapters combine to give precise definitions of what it means for a transformer to "reason," "remember," and be "controlled." These are not just philosophical labels—they have measurable signatures that you can compute and test.

8.1 Defining reasoning geometrically

We all have intuitions about what "reasoning" means, but we need a precise definition to measure it. Here is ours:

Key Idea

A transformer is **reasoning** at layer l and token t when three conditions hold simultaneously:

1. **Interaction**: The curvature $\|\Omega(l,t)\|$ is significantly above baseline—the model is doing non-trivial, nonlinear processing.
2. **Propagation**: The metric factor $P^{(l)}$ has high condition number—the model is

selectively amplifying some directions.

3. **Dependency**: The dependency $D_{l,t}$ is high—the processing at this point actually matters for the output.

All three must be present. Interaction without dependency is "busy work." Dependency without interaction is just passing information through.

8.2 Defining memory

The framework distinguishes two kinds of memory:

Storage memory: the model's weights encode facts ("Paris is the capital of France"). This is a property of the parameters, not the computation on a specific input.

Functional memory: information from early tokens persists and influences the output at later layers. This *is* a property of the computation, and we measure it as **persistent dependency**:

$$\text{Pers}_\tau = \frac{D_\tau}{D_0}$$

If a token from the beginning of the sequence ($\tau = 0$) still has high dependency in the final layers, the model is "remembering" it functionally.

8.3 Defining control

Control means deliberately changing the model's computation by modifying the geometry. The polar decomposition gives us two levers:

- **Metric control**: clamping or modifying the eigenvalues of $P^{(l)}$ to change what the layer amplifies.
- **Rotation control**: modifying $U^{(l)}$ to change the directions of information flow.

The surprise: metric control dominates

The research monograph originally predicted that combining rotation and metric control ("dual control") would work best, with rotation providing the primary lever. The experiments showed the opposite:

What the Data Shows

Metric control alone is sufficient.

Condition	Dep. total	Dep. entropy
Baseline	∼1.00	∼2.45
Rotation-only	∼0.91 (−9%)	∼2.33
Metric-only	∼0.52 (−48%)	∼1.21
Dual (both)	∼0.52 (−48%)	∼1.21

Metric-only and dual control produce *identical* results. Rotation control adds nothing. The reason: the eigenvalue spectrum of $P^{(l)}$ directly controls how gradients flow, while rotation preserves norms and therefore has little effect on gradient magnitudes.

This is an important lesson: the theory predicted one thing, the data showed another, and we updated the theory. Science works.

8.4 Hands-on: running the control experiment yourself

The `ltg` library provides a one-line control experiment:

Listing 8.1: Running the four-condition control study

```python
import ltg

model = ltg.load_model("Qwen/Qwen2.5-7B")

# Run all four control conditions on a single prompt
results = ltg.control_experiment(
    "Explain why the sky appears blue during the day",
    model=model,
    metric_strength=0.5,     # how aggressively to clamp eigenvalues
    rotation_strength=0.3,   # how much to perturb rotation
)

# Print the results
print(f"{'Condition':15s} {'D_total':>8s} {'H_90':>5s} {'Entropy':>8s}")
print("-" * 40)
for name, r in results.items():
    print(f"{name:15s} {r.dep_total:8.3f} {r.dep_horizon_90:5d} "
          f"{r.dep_entropy:8.3f}")
```

```python

# Visualise
ltg.plot_control(results, save_path="ch8_control.png")
```

Try It Yourself

Exercise 8.1 (Verify metric dominance): Run the control experiment on five different prompts (one per task type). For each, compute the percentage drop in D_{total} for metric-only vs. baseline. Is the ∼45% drop consistent across tasks?

Exercise 8.2 (Strength sweep): Vary `metric_strength` from 0.1 to 0.9 in steps of 0.1. Plot D_{total} vs. strength. Is the relationship linear?

Exercise 8.3 (Detecting reasoning): Implement the three-condition reasoning test. A layer is "reasoning" if all three conditions hold:

```python
import numpy as np

result = ltg.analyse(
    "If all mammals are warm-blooded and whales are mammals, "
    "are whales warm-blooded?",
    model=model
)

for l in range(result.n_layers - 1):
    # Condition 1: Curvature above median
    curv_active = result.curvature_by_layer[l] > np.median(
        result.curvature_by_layer)
    # Condition 2: High condition number (selective stretching)
    stretch_active = result.condition_numbers[l] > np.median(
        result.condition_numbers)
    # Condition 3: High dependency
    if result.dependency_profile is not None:
        dep_norm = result.dependency_profile / result.
            dependency_profile.sum()
        dep_active = dep_norm[l] > 1.0 / result.n_layers  # above
             uniform
    else:
        dep_active = False

    if curv_active and stretch_active and dep_active:
        print(f"  Layer {l}: REASONING "
              f"(curv={result.curvature_by_layer[l]:.4f}, "
              f"kappa={result.condition_numbers[l]:.1f}, "
              f"dep={dep_norm[l]:.3f})")
```

Which layers qualify as "reasoning" layers? Are they concentrated at the end of the network?

8.5 Hands-on: measuring functional memory

Functional memory means information from early tokens persists in late layers. We measure this through the dependency map:

Listing 8.2: Measuring functional memory via dependency persistence

```
import ltg
import layer_time_geometry as core
import numpy as np

model = ltg.load_model("Qwen/Qwen2.5-7B")

text = "Paris is the capital of France. What country is Paris in?"

# Get the full (L, T) dependency map using the core library
H_raw = core.extract_hidden_states(model.hf_model, model.tokenizer,
                                   text, model.device)
H_np = H_raw[1:].cpu().numpy()
metric = core.estimate_metric(H_np, n_components=256)
dep_result = core.compute_dependency_density(
    model.hf_model, model.tokenizer, text, metric, device=model.
        device
)

D_lt = dep_result.D_lt  # shape (L, T)
L, T = D_lt.shape

# Persistence: does the first token's dependency persist to late
    layers?
first_token_dep = D_lt[:, 0]  # dependency of token 0 across all
    layers
last_token_dep = D_lt[:, -1]  # dependency of last token across all
    layers

print("Token 0 ('Paris') dependency by layer (first 5, last 5):")
print(f"  Early: {first_token_dep[:5]}")
print(f"  Late:  {first_token_dep[-5:]}")

# Persistence ratio
persistence = first_token_dep[-3:].mean() / (first_token_dep[:3].mean
    () + 1e-12)
```

```
print(f"  Persistence ratio (late/early): {persistence:.3f}")
# > 0.1 suggests the model "remembers" this token
```

Try It Yourself

Exercise 8.4 (Memory decay curve): For a 20-token prompt, compute the persistence ratio for each token. Plot persistence vs. token position. Do earlier tokens have lower or higher persistence? How does this relate to the attention mechanism?

CHAPTER 9

Practical Applications: Diagnosing Model Failures

Companion notebook: `tutorials/ch9_diagnosing_failures.ipynb` — run the code from this chapter interactively.

Why Should I Care?

This chapter shows why all the theory matters. The geometric framework gives you concrete tools to diagnose and sometimes fix specific model failures that you encounter in practice. Every application includes runnable code.

9.1 Application 1: Detecting when a model ignores context

The problem. You provide a document and ask a question about it. The model answers from parametric memory instead of the provided context.

The geometric signature. When the model uses the context, the dependency profile and curvature profile change compared to the no-context version. When the model ignores the context, the two profiles are nearly identical.

Listing 9.1: Detecting context-ignoring with ltg

```
import ltg

```

```python
model = ltg.load_model("Qwen/Qwen2.5-7B")

# Prompt WITH context
r_with = ltg.analyse(
    "According to the following text, the population of Mars Base "
    "Alpha is 847 people. How many people live on Mars Base Alpha?",
    model=model
)

# Prompt WITHOUT context (same question, no document)
r_without = ltg.analyse(
    "How many people live on Mars Base Alpha?",
    model=model
)

# Detect context ignoring
diagnosis = ltg.detect_context_ignoring(r_with, r_without)
print(f"Context influence score: {diagnosis['context_influence']:.2f}")
print(f"Curvature correlation:  {diagnosis['curvature_correlation']:.3f}")
print(f"Dep. entropy diff:      {diagnosis['dependency_entropy_diff']:.3f}")
print(f"Interpretation:         {diagnosis['interpretation']}")

# Compare visually
ltg.compare([r_with, r_without], save_path="ch9_context.png")
```

Try It Yourself

Exercise 9.1 (Context influence sweep): Create prompts where the context provides:

a) Correct, unique information ("The Zorgon treaty was signed in 2019")
b) Correct, common information ("Water boils at 100°C")
c) Contradictory information ("The capital of France is Berlin")

For each, run the context-ignoring detection. In which case does the model use the context most? In which does it ignore it?

9.2 Application 2: Detecting hallucination types

The problem. A model generates plausible-sounding but incorrect text.

The approach. Different hallucination types produce different geometric signatures. We can build a simple classifier:

Listing 9.2: Geometric hallucination diagnostics

```python
import ltg
import numpy as np

model = ltg.load_model("Qwen/Qwen2.5-7B")

def hallucination_check(text, model):
    """Run geometric diagnostics on a prompt."""
    result = ltg.analyse(text, model=model)
    report = ltg.diagnose(result)

    print(f"Prompt: {text[:50]}...")
    print(f"  Peak curvature layer:  {report.curvature_peak_layer}")
    print(f"  Final-3 curv share:    {report.curvature_final3_share
        :.1%}")
    print(f"  Mean condition number: {report.condition_number_mean:.1
        f}")
    if report.dep_entropy:
        print(f"  Dependency entropy:    {report.dep_entropy:.3f}")
    print(f"  Flags:")
    for flag in report.flags:
        print(f"    - {flag}")
    if not report.flags:
        print(f"    (none)")
    return report

# Test on several prompts
prompts = [
    # Likely correct
    "The Earth orbits the Sun once every 365.25 days.",
    # Likely hallucination (unsupported specificity)
    "The first person to climb K2 without oxygen was "
    "Maria Gonzalez in 1987.",
    # Reasoning that might fail
    "If a train travels at 60 mph for 2.5 hours, "
    "it covers exactly 155 miles.",
]

for p in prompts:
    hallucination_check(p, model)
    print()
```

Hallucination type	Geometric signal	What to check
Unsupported addition	High curvature spike at generation point	`curvature_final3_share` > 0.5
Contradiction	Two dependency peaks (conflicting signals)	Multiple local maxima in D_l
Reasoning error	High curvature + low dependency ("busy but useless")	`low_curvature` flag absent, `shallow_dependency` present

Try It Yourself

Exercise 9.2 (Build a detector): Write a function that takes a prompt and returns a hallucination risk score (0–1) based on the geometric diagnostics. Test it on 10 factual claims (5 true, 5 false). Does the score separate them?

Hint: combine curvature concentration, condition number, and dependency entropy into a simple linear score:

```
def risk_score(result):
    curv_conc = result.curvature_by_layer[-3:].sum() / \
                result.curvature_by_layer.sum()
    kappa = result.condition_numbers.max()
    dep_ent = result.dep_entropy or 0
    # Higher concentration + higher selectivity + lower entropy =
        riskier
    return 0.4 * curv_conc + 0.3 * min(kappa / 100, 1) + \
           0.3 * max(0, 1 - dep_ent / 3)
```

9.3 Application 3: Steering model behavior

The problem. You want to change the model's behavior at inference time without retraining.

The framework view. Every steering method is a special case of:

$$\widetilde{H}^{(l)}_{\text{steered}} = U_\Delta \, \widetilde{H}^{(l)} \, P_\Delta$$

The framework tells you *where* to intervene (layers with both high curvature and dependency), *what* to change (the metric factor P, based on our control study), and *how much* (guided by the eigenvalue spectrum).

Listing 9.3: Identifying intervention points for steering

```python
import ltg
import numpy as np

model = ltg.load_model("Qwen/Qwen2.5-7B")
result = ltg.analyse(
    "Write a positive review of the restaurant",
    model=model
)

# Find layers where BOTH curvature and dependency are above median
curv = result.curvature_by_layer
dep = result.dependency_profile[:len(curv)]  # align lengths
curv_thresh = np.median(curv)
dep_thresh = np.median(dep)

print("Recommended intervention layers:")
for l in range(len(curv)):
    if curv[l] > curv_thresh and dep[l] > dep_thresh:
        print(f"  Layer {l}: curv={curv[l]:.4f}, dep={dep[l]:.4f}, "
              f"kappa={result.condition_numbers[l]:.1f}")

# These are the "decision points" where steering has maximum effect
# Intervening at low-curvature or low-dependency layers is wasted
    effort
```

Try It Yourself

Exercise 9.3 (Steering target selection): For three different prompts (factual, creative, analytical), identify the recommended intervention layers. Are they always in the same relative position (e.g., last 20% of layers)?

Exercise 9.4 (Full diagnostic pipeline): Write a function `full_diagnostic(text, model)` that:

1. Runs `ltg.analyse()`
2. Runs `ltg.diagnose()`
3. Identifies intervention layers
4. Prints a formatted report
5. Saves all plots to a directory

This is your personal "model X-ray" tool.

```python
def full_diagnostic(text, model, output_dir="diagnostic"):
    import os; os.makedirs(output_dir, exist_ok=True)

    result = ltg.analyse(text, model=model)
    report = ltg.diagnose(result)

    print(f"=== DIAGNOSTIC REPORT ===")
    print(f"Prompt: {text[:60]}...")
    result.summary()
    print(f"\nFlags:")
    for f in report.flags:
        print(f"  - {f}")

    # Intervention layers
    curv = result.curvature_by_layer
    dep = result.dependency_profile[:len(curv)]
    hot = [(l, curv[l], dep[l]) for l in range(len(curv))
           if curv[l] > np.median(curv) and dep[l] > np.median(
               dep)]
    print(f"\nIntervention targets: layers {[h[0] for h in hot]}"
        )

    result.plot_all(prefix=f"{output_dir}/diag")
    print(f"\nPlots saved to {output_dir}/")
    return result, report

# Usage:
# full_diagnostic("Your prompt here", model)
```

CHAPTER 10

What We Learned and What's Next

10.1 The validated findings

Here is what the framework establishes with cross-model empirical support:

1. **Three-phase computation.** Transformers consistently show early (setup), middle (processing), and late (decision) computational phases, visible in the layer kernel and curvature profile.

2. **Curvature concentrates in final layers.** The most non-trivial computation happens at the end of the network. This replicates across architectures and scales.

3. **Task type is detectable but not dominant.** The type of cognitive task affects dependency metrics, but with modest effect sizes. The geometry does not cleanly separate "arithmetic" from "reasoning."

4. **Reasoning depth is invisible to geometry.** The number of logical steps in a problem has no measurable geometric signature. This was the most important negative result.

5. **Surface features matter.** Prompt length, formatting, and paraphrasing all affect the geometry—you cannot ignore them as confounds.

6. **Metric control dominates rotation control.** For intervening on model computation, modifying the stretching component is sufficient; changing rotation directions adds nothing measurable.

10.2 What this book did not cover

The research monograph contains much more:

- **Geometric algebra**: a unified language for rotation and stretching using bivectors and rotors.
- **Formal proofs**: metric consistency, orthogonal invariance of dependency, local controllability.
- **Detailed worked examples**: six full-length case studies with algorithms (Anti-Vibe, GRASP, RoRA, SPA, hallucination taxonomy, BLADE).
- **Statistical tests**: formal hypothesis testing for curvature localisation, separability, and representation collapse.
- **Diffusion operators and transport theory**: connections to spectral graph theory and parallel transport.

If this book gave you the intuition, the monograph gives you the full mathematical machinery.

10.3 Open problems for you to work on

These are genuine open questions—any of them could be a research project:

1. **Why is reasoning depth invisible?** Our experiments show that step count does not affect geometry. But *something* must differ internally between 1-step and 5-step problems. What is it? Can a nonlinear probe detect what linear geometry cannot?

2. **Cross-architecture geometry.** We tested three models from two families. Does the three-phase structure hold for fundamentally different architectures (Mamba, RWKV, mixture-of-experts)?

3. **Training dynamics.** How does the curvature profile evolve during training? Do the three phases emerge early or late?

4. **Multimodal models.** Vision-language models process images and text together. What does the layer–time geometry look like when two modalities interact?

5. **Better steering.** Given that metric control dominates, can we design a principled, gradient-free steering method that only modifies eigenvalues?

Try It Yourself

Pick one of the open problems above. Design an experiment to make progress on it. What data would you need? What metrics would you compute? Write a one-page proposal.

10.4 The one thing to remember

If you take away just one idea from this book, let it be this:

Transformer computation has measurable geometric structure.

Hidden states form a field on a layer–time grid.
Each layer transition decomposes into rotation and stretching.
Curvature reveals where non-trivial computation occurs.
Dependency reveals which computation matters.

Geometry alone does not explain reasoning.
But geometry plus dependency gives you a principled
framework for understanding, diagnosing, and controlling
what language models do.

Glossary

Term	Plain-language definition
Hidden state $H(l,t)$	The model's internal vector for token t at layer l.
Layer–time grid	The 2D grid with layers on one axis and tokens on the other.
Whitening	PCA-based standardisation that makes dimensions uncorrelated and unit-variance.
Layer kernel K_{layer}	Similarity between two layers, averaged over tokens.
Transition operator $T^{(l)}$	Best linear map from layer l to layer $l+1$.
Polar decomposition	Writing $T = UP$: rotation U times stretching P.
Rotation $U^{(l)}$	The direction-changing part of the layer transition.
Metric factor $P^{(l)}$	The stretching part; eigenvalues show which directions are amplified or compressed.
Condition number κ	Ratio of largest to smallest eigenvalue of P; measures selectivity.
Effective rank	How many directions are actively used (entropy-based).
Curvature $\Omega(l,t)$	Non-commutativity of going "right then up" vs. "up then right" on the grid. Measures nonlinear interaction.
Dependency D_l	Gradient norm of the output score w.r.t. layer l's hidden state. Measures how much layer l matters for the output.
Dependency horizon H_α	Layer at which α% of total dependency is reached.
Functional memory	Early-token information that persists and influences the output in late layers.
Metric control	Modifying the eigenvalues of $P^{(l)}$ to change the model's computation.

Further Reading

- **The research monograph:** Sudjianto and Zhang, *Layer–Time Geometry of Transformer Computation: Kernels, Operators, and Dependency in Language Models* (2026). The complete mathematical treatment with proofs and algorithms.
- **Linear algebra review:** Strang, *Introduction to Linear Algebra* (6th ed.). Chapters on eigenvalues, SVD, and positive-definite matrices are especially relevant.
- **PCA and statistical methods:** Hastie, Tibshirani, and Friedman, *The Elements of Statistical Learning* (2nd ed.). Chapters 3 and 14.
- **Transformer architecture:** Vaswani et al., "Attention Is All You Need" (2017). The original paper.
- **Mechanistic interpretability:** Elhage et al., "A Mathematical Framework for Transformer Circuits" (2021). Complementary to our geometric approach.
- **Experimental design:** Montgomery, *Design and Analysis of Experiments* (10th ed.). Background on the factorial DOE used in Chapter 6.